組別:　1　　組員1(學號姓名):　C108110249　林宇廷　　組員2:　　日期　6/16

實驗12-new：Finite State Machine Application

一、實驗目的：　串列信號產生器及偵測器

學習Verilog code 去design state machine (Mealy machine)

題目: 學習 Verilog code 去 design state machine (digital design Lab 12 and Lab 13)

1. **串列信號產生器:** 請用學號後 8 碼, 模擬的時脈產生電路 其中 1<= 號碼 <=2 的為”1”其它為”0”， 例如: C108110101, 取 08110101 產生 -> 00110101
2. **偵測器:** 學號後 2 碼 mod 4, “餘數” 當 pattern, Ex: 108110101, (01 mod 4) = 01

二、 HDL code and testbench code for the above circuits:

```
// Code your design here
// Lab 12-new: serigen
module lab12new(out,clk, reset,seriout);

//input [3:0] sel;
input clk, reset;
output   seriout;
   output     out;

//input In3;
//input In4;
//output out1;
   reg   [2:0] count; // 8 states, 3-bits
reg   seriout;

   reg       out;
parameter   hight=1,low=0;   // parameter for define the states
parameter   s0=3'd0,s1=3'd1,s2=3'd2;
reg   [2:0]   state;
reg   [2:0]   next_state;

   // 3-bit counter
   //
 always @(posedge clk or negedge reset)
   begin
     if (~ reset)
          count =0;
      else
```

```
        count=count+1;
   end

   // Current sate
always   @(posedge clk or negedge reset)
begin
if   (~reset)
state   =   s0;
else
state   =   next_state;
end


always @(count)
begin
   case (count)   // MUX based combinational ckt
        0:   seriout=0;
        1:   seriout=0;
        2 : seriout=1;
        3 : seriout=1;
        4 : seriout=0;
        5 : seriout=1;
        6 : seriout=0;
        7 : seriout=0;

        default : seriout=0;
        endcase
end


always   @(state   or seriout   )
begin


      case   (state)
        s0: if   (seriout==1 )
                next_state   =   s0;
      else
       next_state   =   s1;
```

```
        s1: if   (seriout==1 )
          next_state   =   s2;
            else next_state   =   s1;
       s2:next_state   =   s0;


       endcase

end
// output stage
//

always   @(state)
begin
case   (state)
s0:   out=low;
s1:   out=low;
s2:   out=hight;


default: out=low;

endcase
end

endmodule================== Testbench==================
// Code your testbench here
// or browse Examples
// Verilog test

`timescale 1ns / 1ns

module lab12newtestbench;

// Inputs
    reg clk, reset;
//
// output
    wire seriout;
```

```
    wire      out;



// Bidirs
    lab12new      u1(out,clk, reset,seriout);


// Initialize Inputs
  initial
clk=0;
always
#5 clk=~ clk;


initial
begin
reset=0;
#20 reset=1;
#400    $finish;

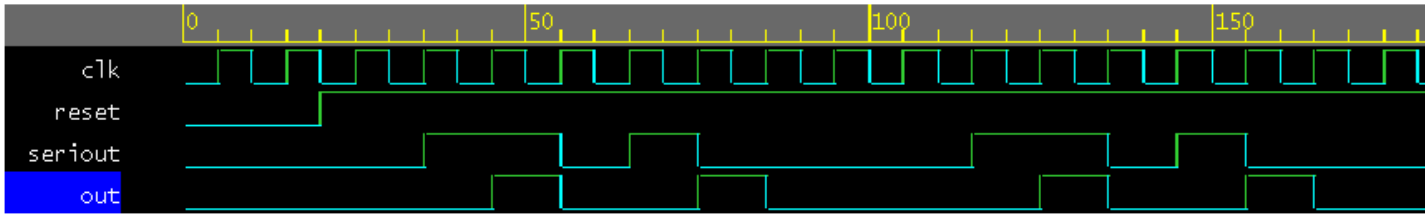
end
initial begin

        $dumpfile("dump.vcd");
        $dumpvars(1);
    end

endmodule
```

三、Simulation Waveform in EDAPlayground

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學號後兩碼 49/4=1 =>01

偵測到串列輸出 01 後 out 出現 1